

Tomas Pereira - Data Scientist

❖ tomas.tp.pereira@gmail.com ❖ www.linkedin.com/in/tomas-tp-pereira/ ❖

PROFILE

As a Computer Science graduate specializing in AI, I am passionate and skilled in the research and development of machine learning models and applications. Inquisitive and thorough, I thrive on understanding systems at a fundamental level with a recent interest in Conversational AI and AI agents that perform tasks independently. I strive to merge my technical skills with my forward-thinking attitude to push the boundaries on what AI can achieve and create innovative tools that solve real-world problems.

WORK EXPERIENCE

Atlas Analytics Lab

Master's/NSERC Researcher

May. 2023– Dec 2024

Concordia University, Montreal, QC

Early Member of the Research Lab, which is Focused on the Development of Deep Learning applications for Computational Pathology. Worked on the Research and Development of a Multimodal Vision-Language Model for Automatic Report Generation using the CLIP Framework and of a Foundational Model for Breast and Colon Cancer Histopathology Tissue.

- Recipient of the NSERC-USRA Research Award in May 2023
- Developed a Framework for Individual Histopathology Object Extraction using Meta's SAM Model
- Interdisciplinary Collaboration with Domain Experts in the Field of Pathology from the CHUM Hospital

Concordia University

Teaching Assistant

Sept. 2023 - Present

Montreal, QC

Teaching Assistant for the Machine Learning Course for both Graduate and Undergraduate Students across Multiple Semesters. Prepared Material and Held Weekly Labs to Teach the Implementation of Machine Learning.

- Developed Teaching Expertise in ML Theory, Pytorch, Numpy and Scikit-Learn
- Guided and Mentored Students in the Development of Their Own Deep Learning Projects
- Tasked with Grading for Theoretical and Practical Assignments, along with Examinations

Bureau Veritas

Systems Specialist

July. 2022 – May. 2023

Montreal, QC

Specialist of Digital Operation of Tasks in the Laboratory and Primary Contact for Data Related Tasks and Issues.

- Successful Refactoring of Datasheets and VBA Macros used in Daily Operations of Lab Analysts
- Performed Data Wrangling and Cleaning Tasks to Half the Time Required by QA Analysts to Vet Records
- Key Figure in Development and Adoption of a New Inventory System for Lab Supplies and Bio Samples

EDUCATION

Concordia University

Master of Computer Science (MAppCompSc)

Sept. 2023 - Present

Montreal, QC

- Specialization in AI, LLMs, Reactive Programming, and Client-Server Applications
- Recipient of the Concordia Merit Scholarship Award (2024 and 2025)
- Studied Representation Learning and Reinforcement Learning at MILA - Quebec AI Institute
- Current Project in Conversational AI for a Multi-Speaker Text-to-Speech Model

Concordia University

Bachelor of Computer Science (BCompSc)

Sept 2019 - May 2023

Montreal, QC

- Recipient of the Government of Quebec Award under the Merit Scholarship Program (2021)

PROJECTS

Research Project - Updating Method-Level Comments Using Generative AI

This Project Tackled the Application of Generative AI for the Domain of Software Engineering. Two Models were developed with the Task of Updating Existing Code-Comments when the Relevant Code is Changed.

- Led the Effort to Fine Tune the CodeT5 Transformer and Google's Gemma (LLM) Model
- Studied the Effectiveness of Varied Inputs and Gradual Information Gain with Common Textual Similarity Benchmarks like BLEU and Meteor Score

Research Project - Reinforcement Learning with Multi Instance Learning For Efficient WSI Analysis

Using Reinforcement Learning, this Project developed an Agent which could Detect the Tissue Segments on a Whole Slide Image more Efficiently than other Existing Pre-processing Methods

- Early work in the Application of RL for Computational Pathology, which is Mostly Unexplored for the Field
- Future Extension of this work is Aimed at Developing a Cancer Classifier Which Can Perform as Well as the State-of-the-art While Utilization Significantly Less Compute Power

PLAY Framework Reactive System

This Project Developed a Reactive Application Interfacing the YoutubeV3 API using Asynchronous Message Passing. The Aim was to Learn the Concepts of Reactive Programming and Modern Java Development

- Led Group and Assigned Tasks to Team Members To Ensure Steady And Efficient Development
- Developed a Component for Sentiment Analysis Based on Video Descriptions

Shipping Carrier Recommendation System

Early Project in my Machine Learning Path in Which we Built a Simple Multiclass Classifier using Random Forest which was adopted in an active Logistics Startup to Expedite the Selection of Trucking Service per Shipment.

- Led to a Significant Reduction of Time Spent by Dispatch during Selection.

SKILLS & INTERESTS

- **Technical Skills:**
 - o Languages: Python, Java, SQL, VBA, C++, Ruby, C#
 - o Technologies: Pytorch, Numpy, SKLearn, OpenAI API, HuggingFace, PLAY, SpeechBrain, OpenCV, Gymnasium, Pandas, SLURM
 - o Other: Deep Learning, Neural Networks, R&D, Computer Vision, NLP, LLMs, Decision-Making Systems, Medical Imaging, Data Science, Backend Programming
- **Soft Skills:**
 - o Problem Solving; Critical Thinking; Collaboration; Leadership; Communication; Analysis; Mentoring; Presentation; Innovation; Project Management; Attention to Detail; Adaptability; Creativity;
 - o Spoken Languages: English, French and Portuguese (Trilingual)
- **Interests:** Conversational AI; Dungeons and Dragons; Game Development; Gardening; Running; Rugby;

REFERENCES: Available Upon Request